

# Designing and Analysis of Algorithms

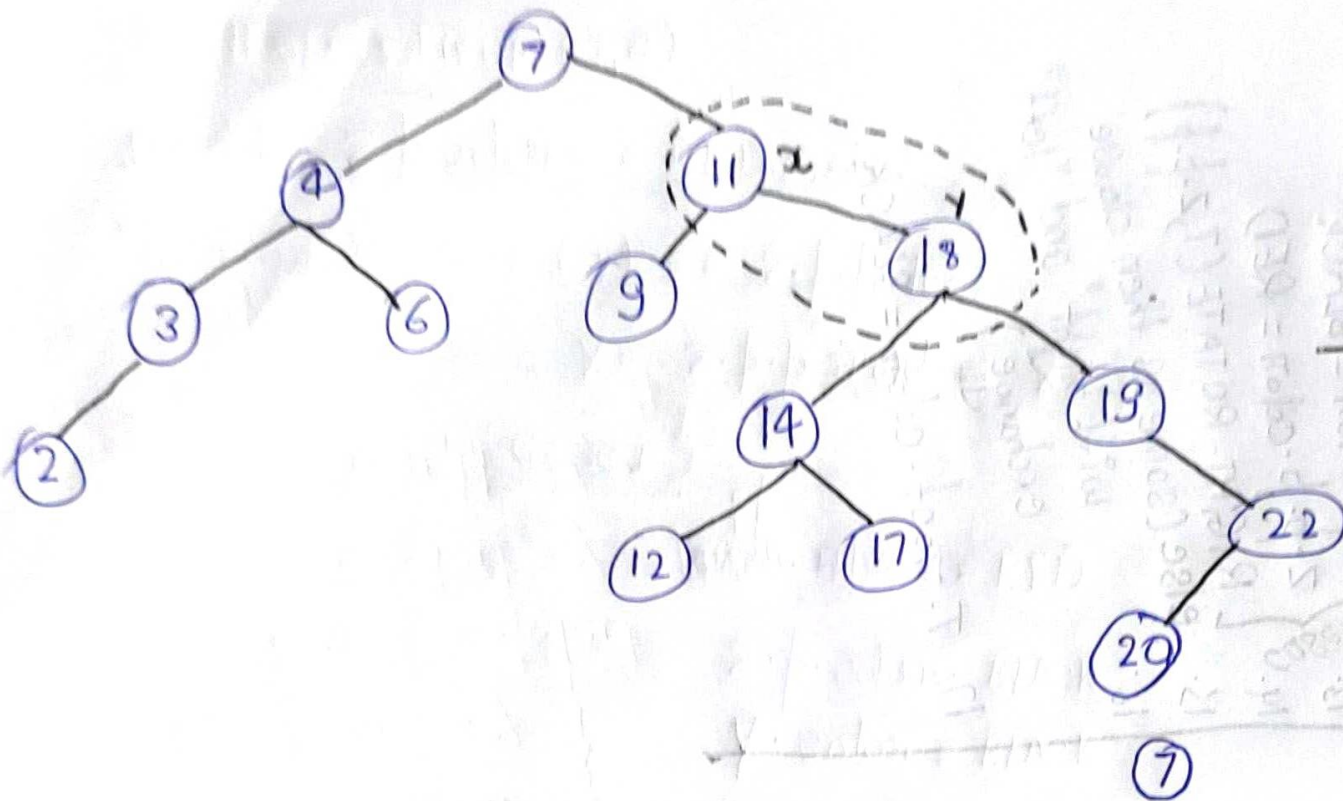
Course Code: ECS 5101/CS514

- Dr Rahul Mishra
- IIT Patna

- **Lecture 21**

## LEFT-ROTATE( $T.x$ )

1.  $y = x.right$  // set  $y$
2.  $x.right = y.left$  // set  $y$
3. if  $y.left \neq T.nil$  // turn  $y$ 's left subtree into  $x$ 's right subtree
4.  $y.left.p = x$
5.  $y.p = x.p$  // link  $x$ 's parent to  $y$
6. if  $x.p == T.nil$
7.  $T.root = y$
8. else if  $x == x.p.left$
9.  $x.p.left = y$
10. else  $x.p.right = y$
11.  $y.left = x$  // put  $x$  on  $y$ 's left
12.  $x.p = y$



Left rotate

①  
 $x = 11$   
 $y = x \cdot \text{right} = 18$   
 if  $y \cdot \text{left} \neq T \cdot \text{nil}$   
     ( $y \cdot \text{left} = 14$ )  
      $y \cdot \text{left} \cdot p = x = 11$   
      $y \cdot p = x \cdot p = 7$   
 if  $x \cdot p == T \cdot \text{nil}$   
     No true  
 elseif  $x == x \cdot p \cdot \text{left}$   
     No true  
 else  $x \cdot p \cdot \text{right} = y$   
      $y \cdot \text{left} = x$   
      $x \cdot p = y$

# \* Graphs and Their Applications :-

## ► Graphs and Multigraphs :-

A graph  $G$  consists of two things:

- (1) A set  $V$  of elements called nodes (or points or vertices)
- (2) A set  $E$  of edges such that each edge  $e$  in  $E$  is identified with a unique (unordered) pair  $[u, v]$  of nodes in  $V$ , denoted by  $e = [u, v]$ .

↳ some times, we also write  $G = (V, E)$ .

⇒ Suppose  $e = [u, v]$ . Then the node  $u$  and  $v$  are called endpoints of  $e$ , and  $u$  and  $v$  are said to be adjacent nodes or neighbors.

⇒ The degree of a node  $u$ , written  $\deg(u)$  is the number of edges containing  $u$ .

⇒ IF  $\deg(u) = 0 \Rightarrow$  if  $u$  does not belong to any edge ( $u$  is called isolated node)

\* A path  $P$  of length  $n$  from a node  $u$  to a node  $v$  is defined as a sequence of  $n+1$  nodes.

$$P = (v_0, v_1, v_2, \dots, v_n)$$

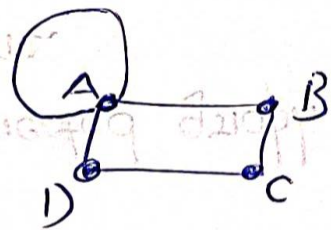
\* A Graph  $G$  is said to be connected if there is a path between any two of its nodes.

\* A graph  $G$  is connected if and only if there is a simple path between any two nodes in  $G$ .

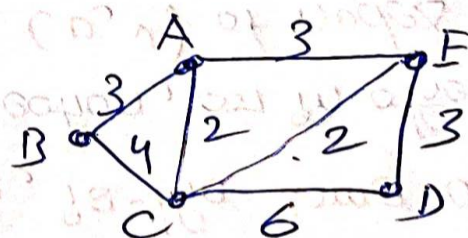
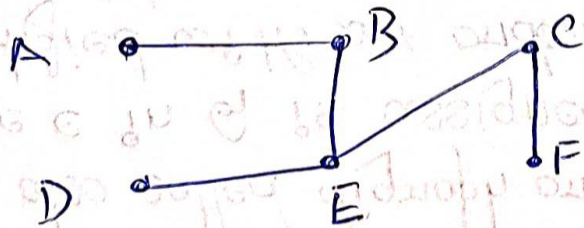
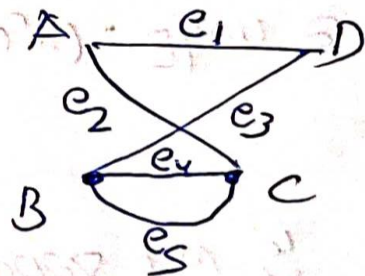
\* A Graph  $G$  is said to be complete if every node  $u$  in  $G$  is adjacent to every other node  $v$  in  $G$ .

- \* A complete graph with  $n$  nodes will have  $n(n-1)/2$  edges.
- \* A connected graph  $T$  without any cycles is called tree graph or free tree or simply a tree.
- \* If  $T$  is a finite tree with  $n$  nodes, then  $T$  will have  $n-1$  edges.
- \* A Graph  $G$  is said to be labeled if its edges are assigned data.

\* Loop :-



\* Multiedges :-



## Directed Graphs :-

④

A directed Graph  $G$ , also called **digraph** or **graph**, is the same as a multigraph except that each edge  $e$  in  $G$  is assigned a direction, or in other words, each edge  $e$  is identified with an ordered pair  $(u, v)$  of nodes in  $G$  rather than unordered pair  $[u, v]$ .

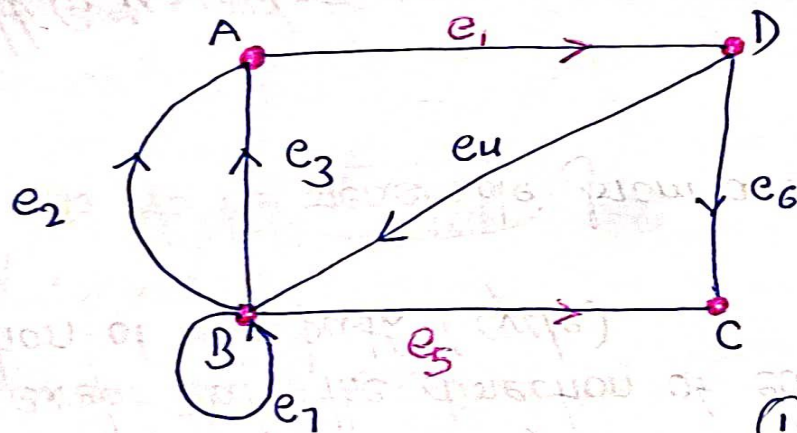
Suppose  $G$  is a directed graph with a directed edge  $e = (u, v)$ . Then  $e$  is also called an arc.

- (1)  $e$  begins at  $u$  and ends at  $v$ .
- (2)  $u$  is the origin or initial point of  $e$ , and  $v$  is the destination or terminal point of  $e$ .
- (3)  $u$  is a predecessor of  $v$ , and  $v$  is a successor or neighbor of  $u$ .
- (4)  $u$  is adjacent to  $v$ , and  $v$  is adjacent to  $u$ .

- \* The outdegree of a node  $u$  in  $G$ , written  $\text{outdeg}(u)$ , is the number of edges beginning at  $u$ .
- \* Similarly, the indegree of  $u$ , written  $\text{indeg}(u)$ , is the number of edges ending at  $u$ .
- \* A node  $u$  is called a source if it has a positive outdegree but zero indegree.
- \* A node  $u$  is called a sink if it has a zero outdegree but a positive indegree.
- \* The notions of path, simple path, and cycle carry over from undirected graphs to directed graphs except that the direction of each edge in a path (cycle) must agree with the direction of the path (cycle).
- \* A node  $v$  is said to be reachable from a node  $u$  if there is a (directed) path from  $u$  to  $v$ .
- \* A directed graph  $G$  is said to be connected or strongly connected, if for each pair  $u, v$  of nodes in  $G$  there is a path from  $u$  to  $v$  and there is also a path from  $v$  to  $u$ .

\*  $G$  is said to be *unilaterally connected* if any pair  $u, v$  of nodes in  $G$  there is a path from  $\boxed{u \text{ to } v}$  or a path from  $\boxed{v \text{ to } u}$ .

⑥



A directed graph  $G$  with 4 nodes and 7 (directed) edges.

- (iv)  $P_1 = (D, B, A)$  is a path
- (v)  $G$  is unilaterally connected.
- (vi)  $\text{indeg}(D) = 1$
- (vii)  $\text{outdeg}(D) = 2$
- (viii)  $C$  is a sink
- (ix) No node in  $G$  is source.

- (i) edges  $e_2$  and  $e_3$  are parallel (begin B end A)
- (ii)  $e_7$  is a loop: starts and ends at same point
- (iii) The sequence  $P_1 = (D, C, B, A)$  is not a path, since  $(C, B)$  is not an edge. i.e; the direction of edge  $e_5 = (B, C)$  does not agree with direction

## More :- Terminology ▶

⑦

- \* Let  $T$  be any non empty tree graph.
- \*  $T$  with designated node  $R$  is called a **rooted tree** and  $R$  is called its root.
- \* A directed graph  $G$  is said to be **simple** if  $G$  has no parallel edges.
- \* A **simple graph  $G$**  may have loops, but it cannot have more than one loop at a given node.
- \* A **non directed Graph  $G$**  may be viewed as a simple directed graph by assuming the each edge in  $G$  represents two directed edges.

Note: Unless otherwise stated or implied, the term "**graph**" shall mean simple directed graph, and the term "**edge**" shall mean directed edge.

## Sequential Representation of Graphs →

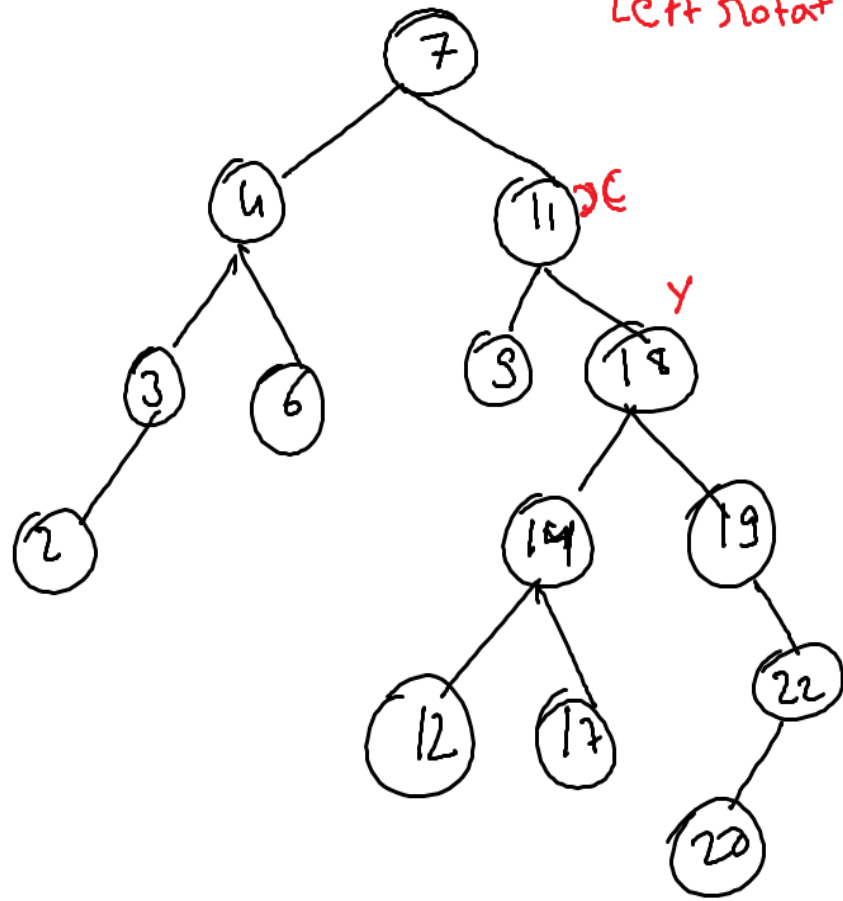
- \* Sequential representation of  $G$  is managed by adjacency matrix.

## Adjacency Matrix ÷

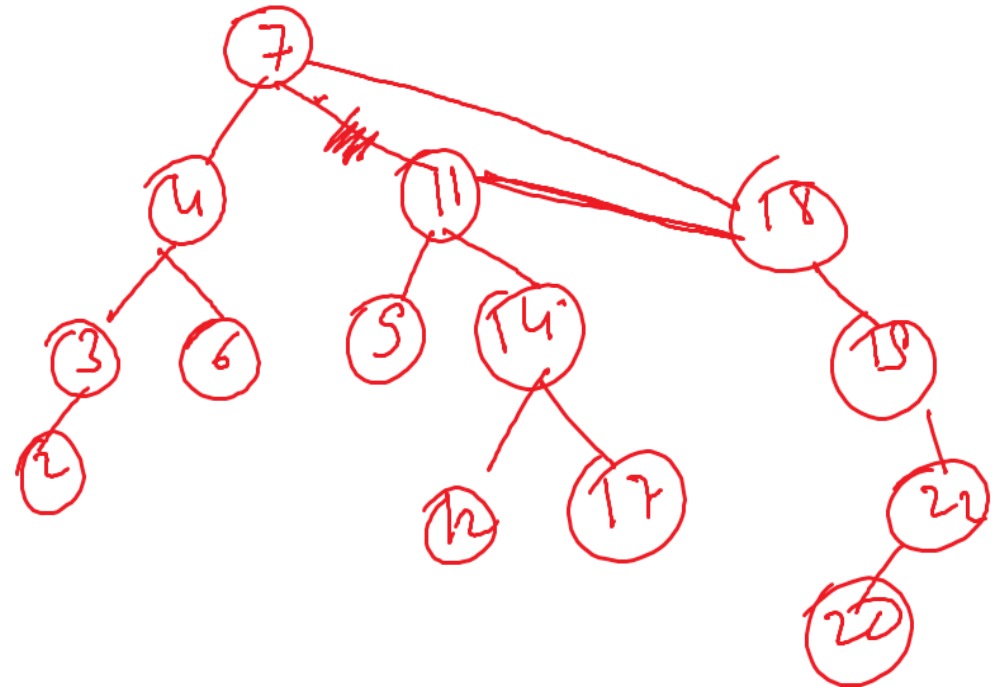
- Suppose  $G$  is a simple directed graph with  $m$  nodes, and suppose the nodes of  $G$  have been ordered and are called  $v_1, v_2, \dots, v_m$ . The adjacency matrix  $A = (a_{ij})$  of the graph  $G$  is the  $m \times m$  matrix defined as follows:

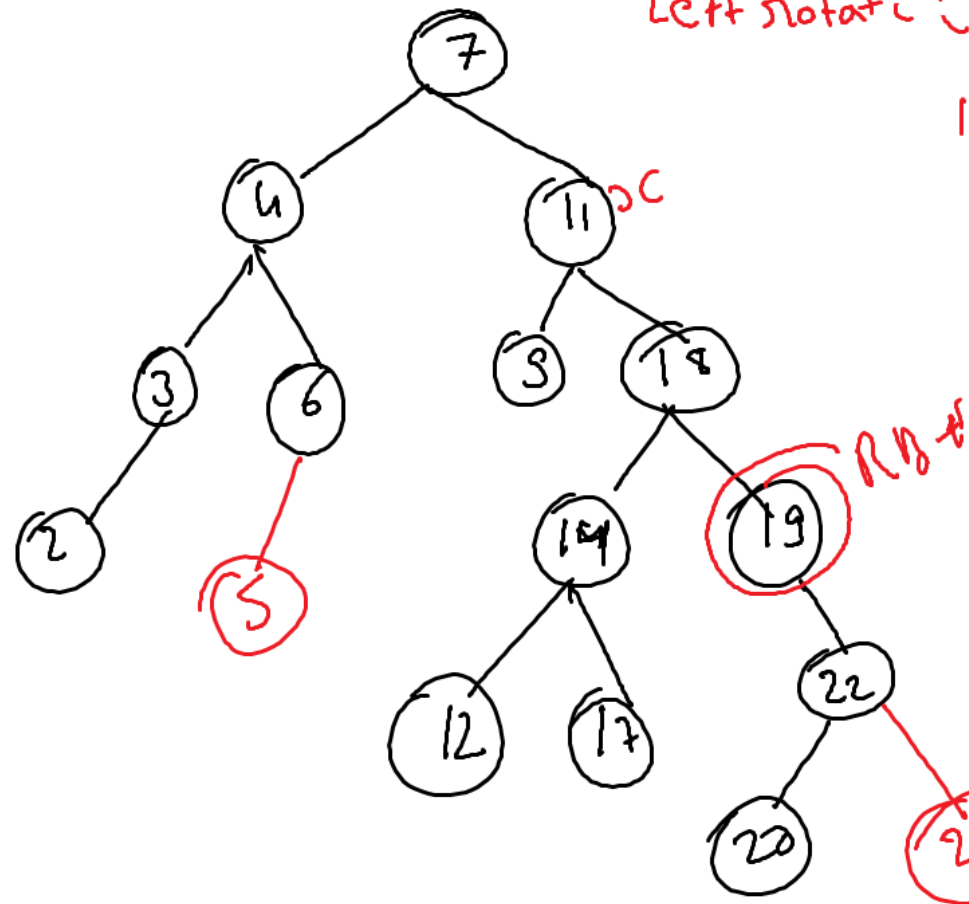
$$a_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is adjacent } v_j, \text{ i.e., if there is an edge } (v_i, v_j) \\ 0 & \text{otherwise} \end{cases}$$

- Such a matrix  $A$ , which contains entries of only 0 and 1, is called a bit matrix or a Boolean matrix.
- The adjacency matrix  $A$  of the graph  $G$  does depend on the ordering of the nodes of  $G$ , i.e., a different ordering of the nodes may result in a different adjacency matrix.
- The matrices resulting from two different orderings are closely related in that one can be obtained from the other by simply interchanging rows and columns.



- ①  $y = x.\text{right}$
- ②  $x.\text{right} = y.\text{left}$





Left rotate (T. 11)

Insertion in RB

~~S~~ (RB)

RB Fixup :- maintain the property of RB  
 → will follow BST insertion & RB fix up properties  
 ② RB - Deficit

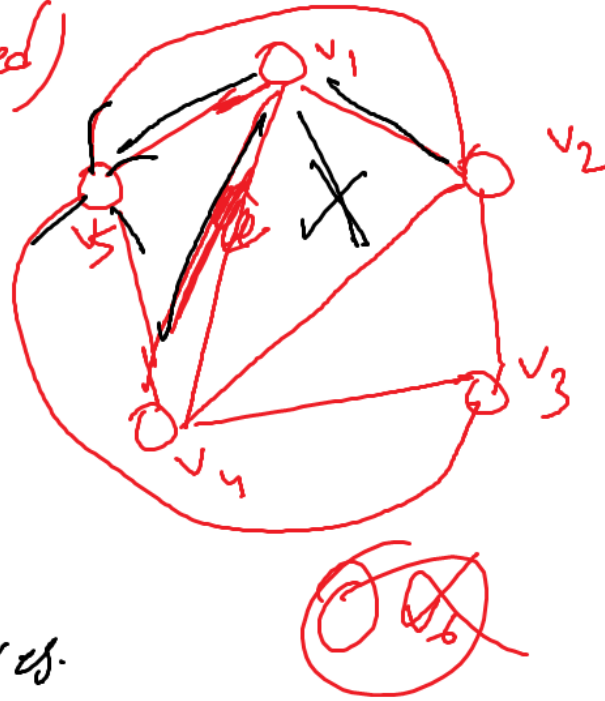
→ \* In RB tree Insertion & Deletion will follow the BST. Fix up

~~①~~ degree of vertex (undirected)

② connected Any pair

③ Complete

- All the vertices will be connected to all other vertices.
- All the vertices are adjacent



$$\deg(v_1) = 3$$

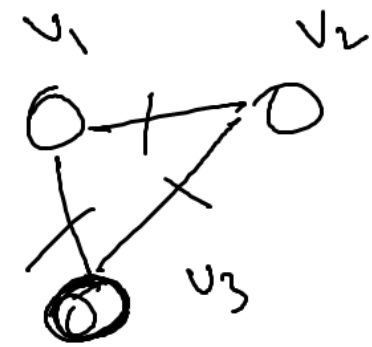
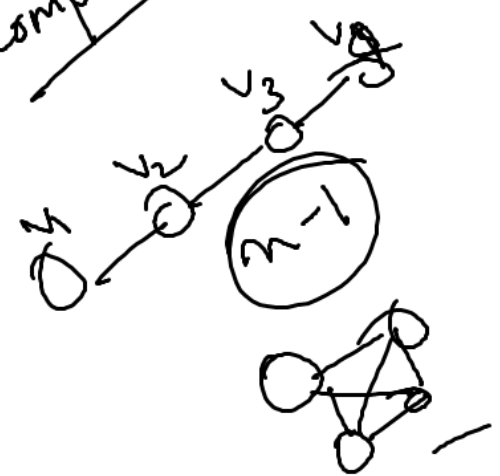
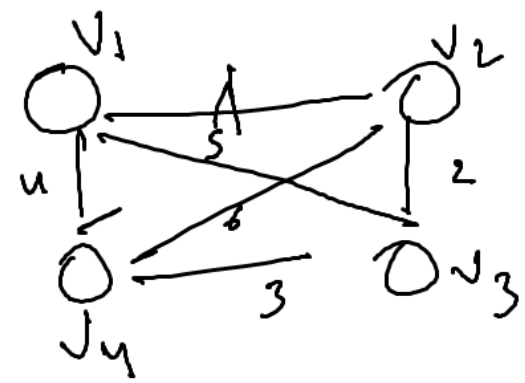
$$\deg(v_2) = 4$$

$$\deg(v_3) = 3$$

$$\deg(v_4) = 4$$

$$\deg(v_5) = 4$$

Connected  
Complete

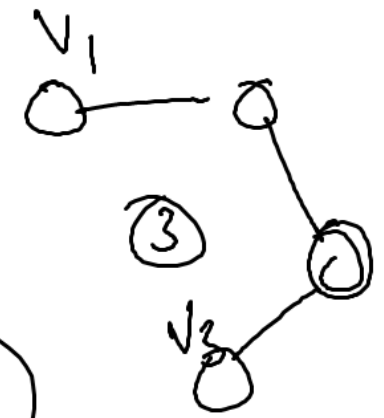


• This is complete graph

$$\frac{n(n-1)}{2} \Rightarrow n \text{ vertices}$$

$$e = \frac{4(3)}{2} = 6$$

$$\frac{n(n-1)}{2} = \frac{3(3-1)}{2} = 3$$



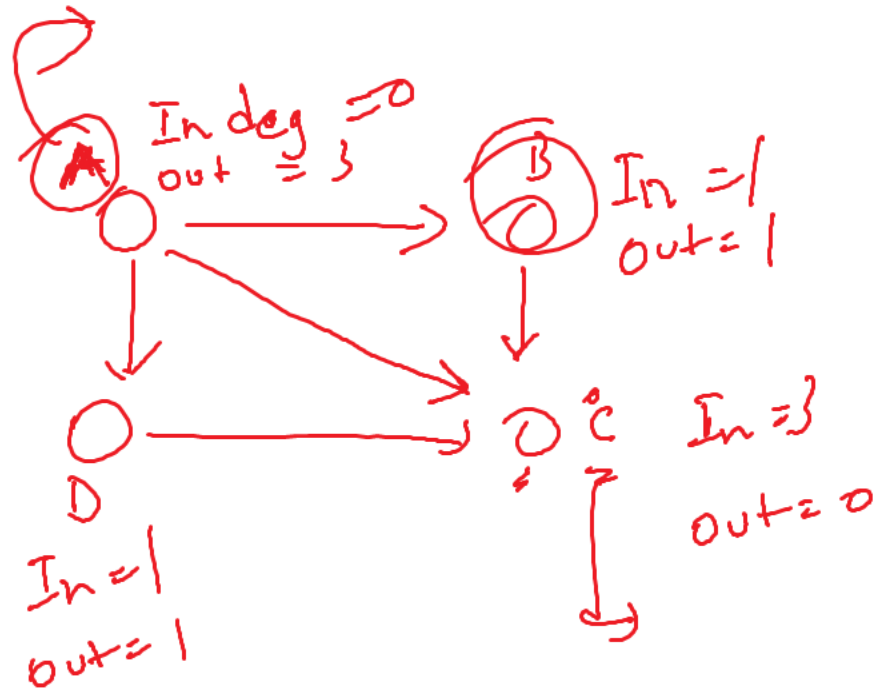
$$\begin{matrix} n-1 \\ \downarrow \\ 2 \end{matrix}$$

In degree

out degree

reachability

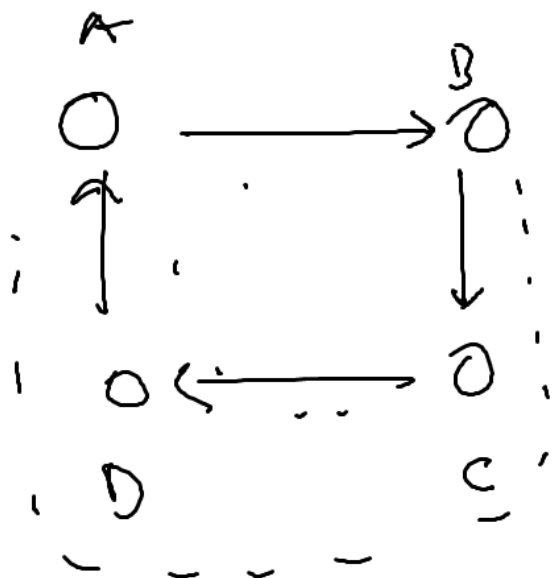
degree



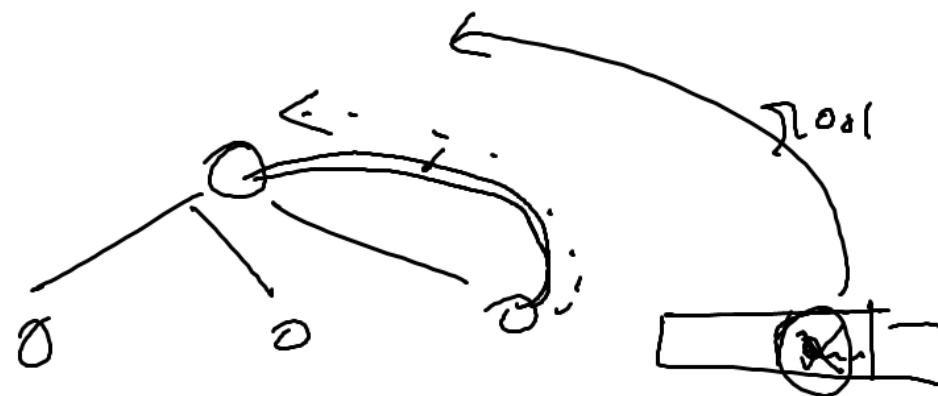
Source  $\rightarrow$  out = +ve  
In = 0

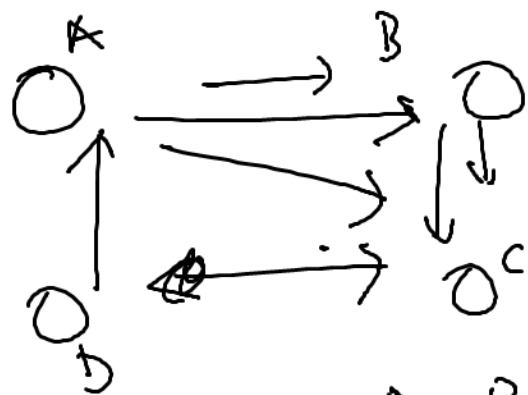
Sink  $\rightarrow$  out = 0  
In = +ve

- From A there is path to all other nodes



connected graph





$$\begin{matrix} & A & B \\ A & 1 & 1 \\ B & & \\ C & & \\ D & & \end{matrix}$$

$$\Rightarrow \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} =$$

$$\begin{matrix} C & D \\ 0 & 0 \end{matrix}$$

$$\begin{matrix} & A & B & C & D \\ A & 1 & 1 & 1 & 1 \\ B & 1 & 1 & 1 & 1 \\ C & 0 & 0 & 1 & 0 \\ D & 0 & 0 & 0 & 0 \end{matrix}$$